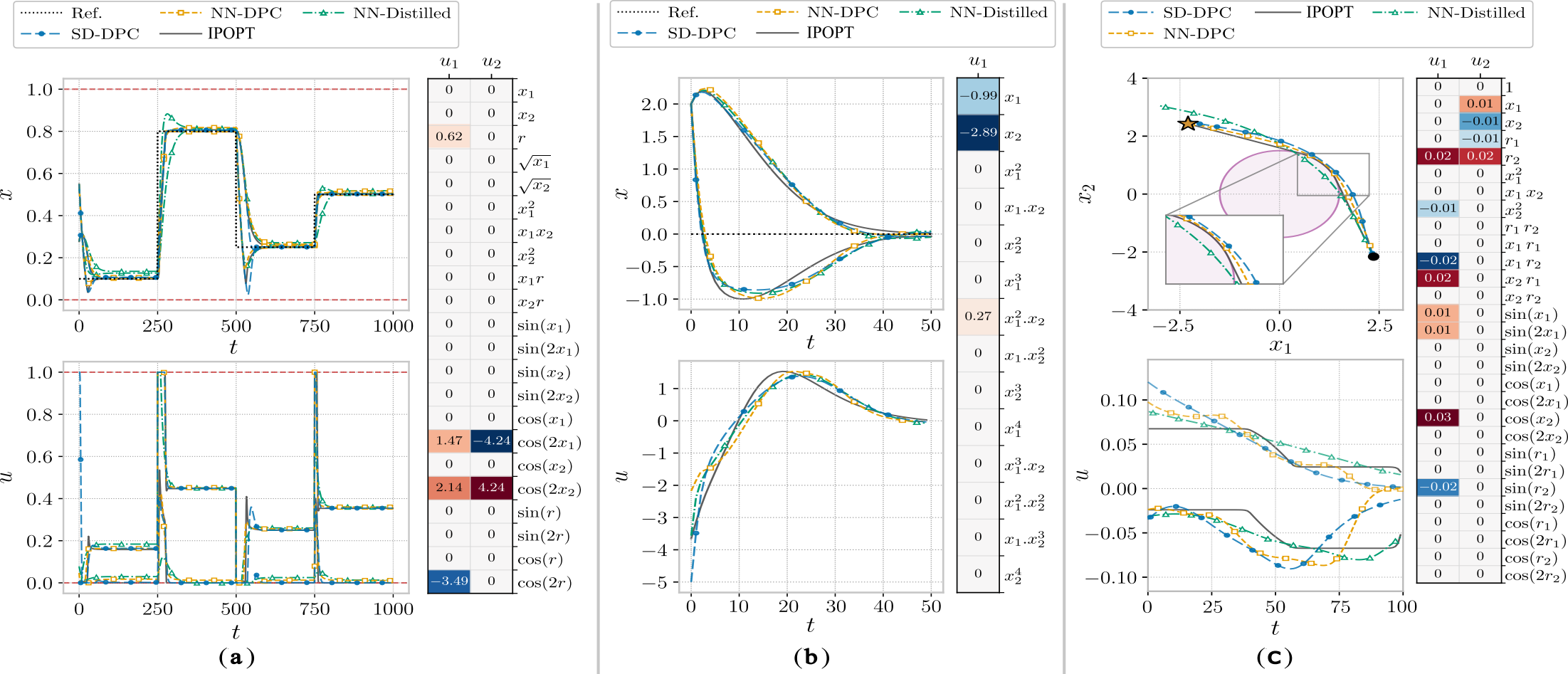

Ref.
SD-DPC
NN-DPC
IPOPT
NN-Distilled
$x$
$u$
$t$
$u_1$ $u_2$
0 0 $x_1$
0 0 $x_2$
0.62 0 $r$
0 0 $\sqrt{x_1}$
0 0 $\sqrt{x_2}$
0 0 $x_1^2$
0 0 $x_1x_2$
0 0 $x_2^2$
0 0 $x_1r$
0 0 $x_2r$
0 0 $\sin(x_1)$
0 0 $\sin(2x_1)$
0 0 $\sin(x_2)$
0 0 $\sin(2x_2)$
0 0 $\cos(x_1)$
1.47 −4.24 $\cos(2x_1)$
0 0 $\cos(x_2)$
2.14 4.24 $\cos(2x_2)$
0 0 $\sin(r)$
0 0 $\sin(2r)$
0 0 $\cos(r)$
−3.49 0 $\cos(2r)$
(a)
$u_1$
−0.99 $x_1$
−2.89 $x_2$
0 $x_1^2$
0 $x_1.x_2$
0 $x_2^2$
0 $x_1^3$
0.27 $x_1^2.x_2$
0 $x_1.x_2^2$
0 $x_2^3$
0 $x_1^4$
0 $x_1^3.x_2$
0 $x_1^2.x_2^2$
0 $x_1.x_2^3$
0 $x_2^4$
(b)
$x_2$
$x_1$
$u_1$ $u_2$
0 0 1
0 0.01 $x_1$
0 −0.01 $x_2$
0 −0.01 $r_1$
0.02 0.02 $r_2$
0 0 $x_1^2$
0 0 $x_1\ x_2$
−0.01 0 $x_2^2$
0 0 $r_1\ r_2$
0 0 $x_1\ r_1$
−0.02 0 $x_1\ r_2$
0.02 0 $x_2\ r_1$
0 0 $x_2\ r_2$
0.01 0 $\sin(x_1)$
0.01 0 $\sin(2x_1)$
0 0 $\sin(x_2)$
0 0 $\sin(2x_2)$
0 0 $\cos(x_1)$
0 0 $\cos(2x_1)$
0.03 0 $\cos(x_2)$
0 0 $\cos(2x_2)$
0 0 $\sin(r_1)$
0 0 $\sin(2r_1)$
−0.02 0 $\sin(r_2)$
0 0 $\sin(2r_2)$
0 0 $\cos(r_1)$
0 0 $\cos(2r_1)$
0 0 $\cos(r_2)$
0 0 $\cos(2r_2)$
(c)